

Circle Theorems

Seven theorems, one habit: spot the shape, then name the theorem next to every angle.



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eliteigcse.com

THE TOPIC ON ONE MAP

Circle Theorems

01 Angles in circles

centre = $2 \times$ circumference

same segment

cyclic quadrilateral

02 Tangents

tangent $\perp$ radius

two tangents equal

alternate segment

03 Chords and secants

$AP \cdot PB = CP \cdot PD$

$PA \cdot PB = PC^2$

04 Proofs

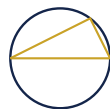
similar triangles

prove parallel

THE SEVEN THEOREMS See the shape, name the theorem:



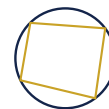
centre = $2 \times$
circumference



semicircle
= 90°



same segment
equal



cyclic quad
opposite = 180°



tangent $\perp$
radius



two tangents
equal



alternate
segment

01 Angles in Circles

Find the triangles made with the centre — they are isosceles (two radii). Then use: centre = $2 \times$ circumference, semicircle = 90° , same segment equal, cyclic quadrilateral opposite angles 180° .

WHEN YOU SEE → YOU DO

O the centre, lines to O

→ isosceles triangle (two radii)

angle at O and at the edge on the same arc

→ centre = $2 \times$ circumference

“AOC is a diameter”

→ angle in a semicircle = 90°

two angles from the same chord

→ angles in the same segment are equal

four points on the circle, joined

→ opposite angles add to 180°

“give a reason for each stage”

→ the full theorem, every time

WORKED MODEL • NOVEMBER 2024 P2H

$ACD = 54^\circ$, $ODC = 28^\circ$. Find AOD , CAO and ABC .

1 $AOD = 2 \times 54 = 108^\circ$

centre = $2 \times$ circumference

2 $OCD = 28^\circ$, so $OCA = 26^\circ = CAO$

isosceles (radii)

3 $ADC = 36 + 28 = 64^\circ$

$$ODA = \frac{180 - 108}{2}$$

ANSWER $ABC = 180 - 64 = 116^\circ$

IDEA 01

Practise: Angles in Circles

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Centre and circumference

CH 4 · TOPIC 4 2 marks

O is the centre. Angle $ABC = 47^\circ$ at the circumference. Find angle AOC and give the reason.

Q2 DR ESLAM Cyclic quadrilateral

CH 4 · TOPIC 4 2 marks

$ABCD$ is a cyclic quadrilateral with angle $A = 73^\circ$. Find angle C and give the reason.

SELECTED PRACTICE

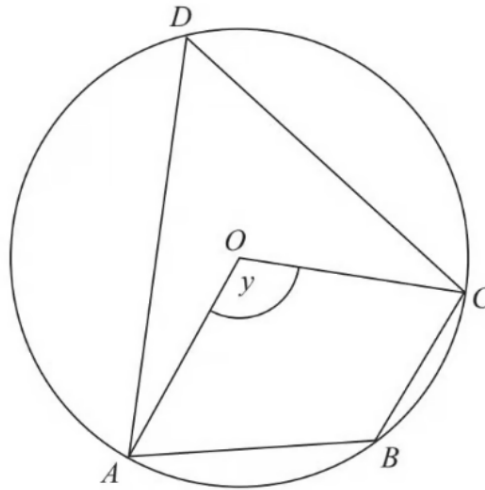
Q3

PRACTICE

Practice · Medium

CH 4 · TOPIC 4 4 marks

4

Diagram **NOT**
accurately drawn

A , B , C and D are points on the circumference of a circle, centre O .

Angle $AOC = y$.

Find the size of angle ABC in terms of y .

Give a reason for each stage of your working.

(4 marks)

Q4

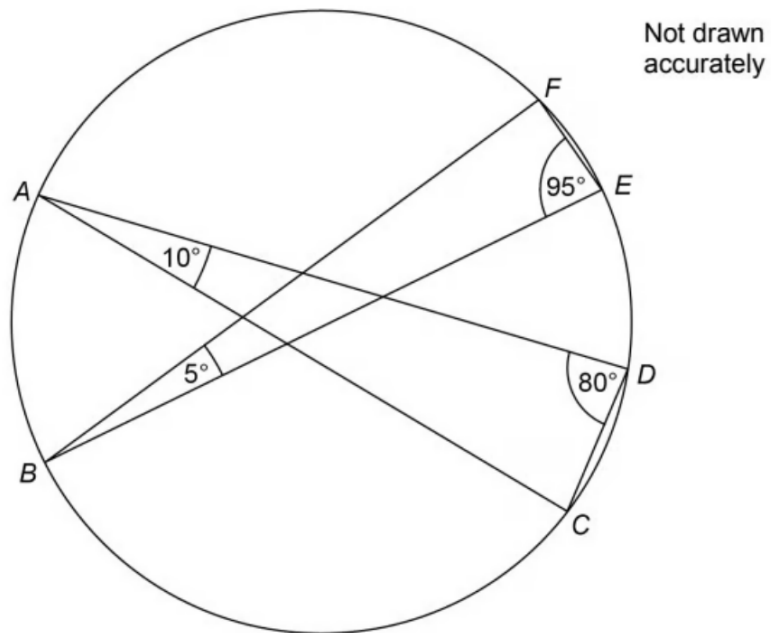
PRACTICE

Practice · Medium

CH 4 · TOPIC 4

2 marks

17 A, B, C, D, E and F are points on a circle.



Circle the line that is a diameter of the circle.

BE	AD	AC	BF
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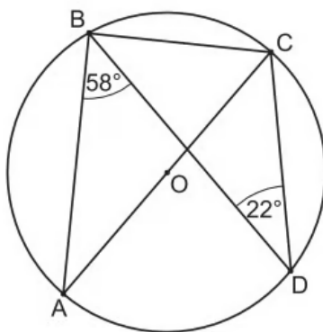
(2 marks)

Q5**PRACTICE**

Practice · Hard

CH 4 · TOPIC 4 5 marks

- 14 (a)** A, B, C and D are points on the circumference of a circle, centre O.
AC is a diameter of the circle.
Angle ABD = 58° .
Angle CDB = 22° .

**Not to scale**

Work out the sizes of angle ACD and ACB, giving reasons for your answers.

Angle ACD = $^\circ$

(2 marks)

(b) Angle ACB = $^\circ$

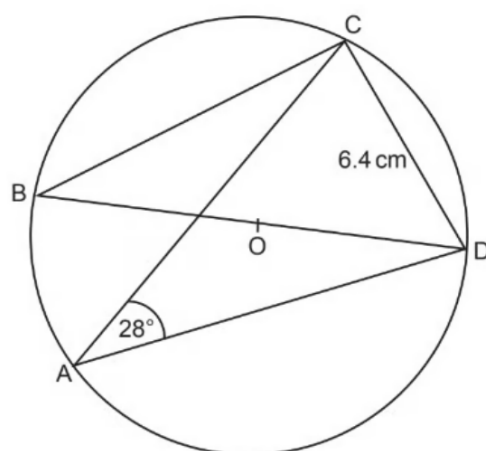
(3 marks)

Q6**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 4 5 marks

- 18 A, B, C and D are points on the circumference of a circle, centre O.



Not to scale

Angle CAD = 28° and CD = 6.4cm.

BD is a diameter of the circle.

Calculate the area of the circle.

..... cm^2 **(5 marks)**

PAST PAPERS

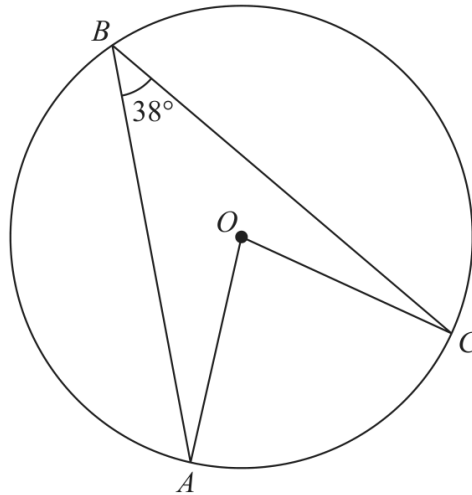
Q7

PAST PAPER

Edexcel 4MA1 · January 2020 · Paper 1HR · Q14

CH 4 · TOPIC 4 4 marks

14

Diagram **NOT**
accurately drawn

A , B and C are points on a circle, centre O .
Angle $ABC = 38^\circ$

Work out the size of angle OAC .
Give a reason for each stage of your working.

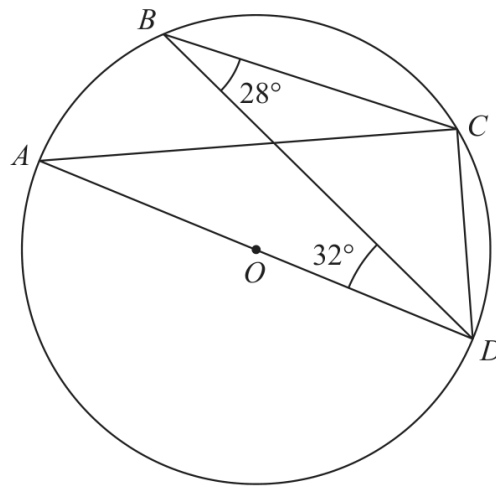
○

(Total for Question 14 is 4 marks)

Q8 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2H · Q13

CH 4 · TOPIC 4 4 marks

13

Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle, centre O .
 AOD is a diameter of the circle.

Angle $CBD = 28^\circ$

Angle $BDA = 32^\circ$

Find the size of angle BDC .

Give a reason for each stage of your working.

○

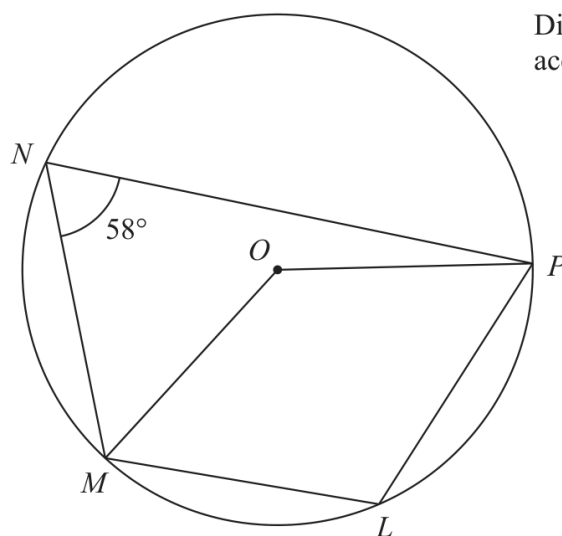
(Total for Question 13 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q16

CH 4 · TOPIC 4 4 marks

16

Diagram **NOT**
accurately drawn L, M, N and P are points on a circle, centre O Angle $MNP = 58^\circ$ (a) (i) Find the size of angle MLP

○

(ii) Give a reason for your answer.

(b) Find the size of the reflex angle MOP

○

(Total for Question 16 is 4 marks)

Q10**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2H · Q13

CH 4 · TOPIC 4 3 marks

13 P, Q, R, S and T are points on a circle with centre O .

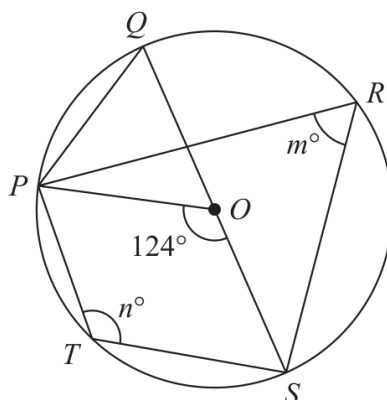


Diagram **NOT**
accurately drawn

QOS is a diameter of the circle.

angle $POS = 124^\circ$ angle $PRS = m^\circ$ angle $PTS = n^\circ$

(a) Find the value of

(i) m

.....

(ii) n

.....

(b) Find the size of angle QPO .

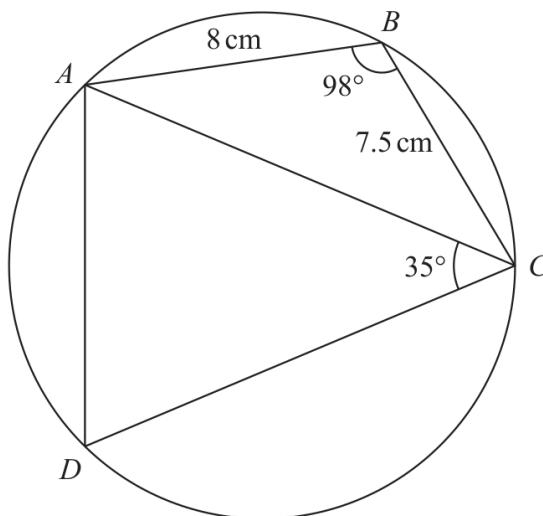
..... °

(Total for Question 13 is 3 marks)

Q11 PAST PAPER Edexcel 4MA1 • May 2021 • Paper 2H • Q18

CH 4 • TOPIC 4 6 marks

18

Diagram **NOT**
accurately drawn

$ABCD$ is a quadrilateral where A , B , C and D are points on a circle.

$$AB = 8\text{ cm}$$

$$BC = 7.5\text{ cm}$$

$$\text{Angle } ABC = 98^\circ$$

$$\text{Angle } ACD = 35^\circ$$

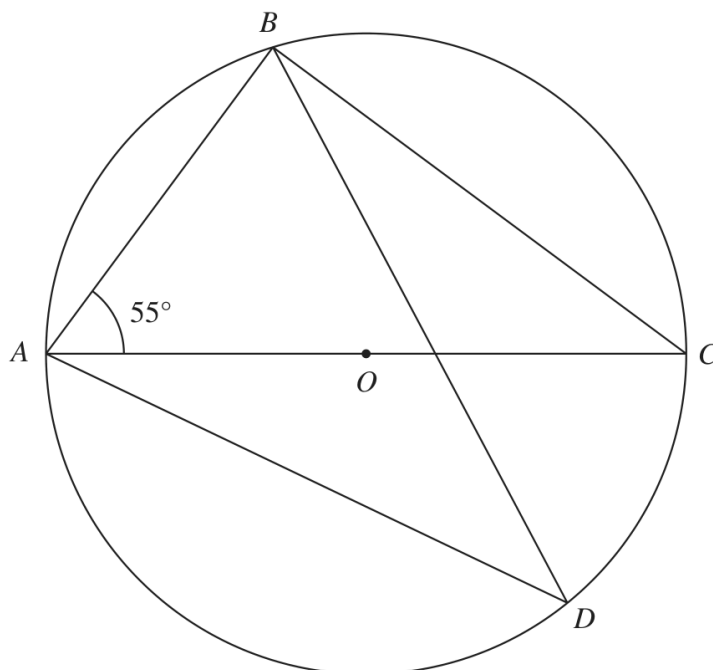
Work out the perimeter of quadrilateral $ABCD$.
Give your answer correct to one decimal place.

..... cm

(Total for Question 18 is 6 marks)

Q12**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q14

CH 4 · TOPIC 4 4 marks**14**Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle, centre O
 AOC is a diameter of the circle.

Angle $BAC = 55^\circ$

Work out the size of angle ADB
Give a reason for each stage of your working.

o

(Total for Question 14 is 4 marks)

Q13**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2H · Q16

CH 4 · TOPIC 4 4 marks

16 D, E, F and G are points on a circle, centre O

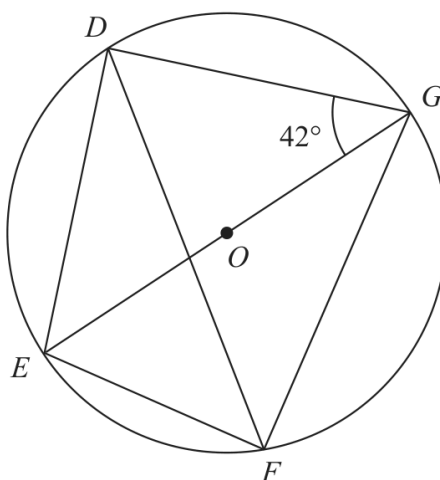


Diagram **NOT**
accurately drawn

EOG is a diameter of the circle.

Angle $EGD = 42^\circ$

Calculate the size of angle DFG

Give a reason for each stage of your working.

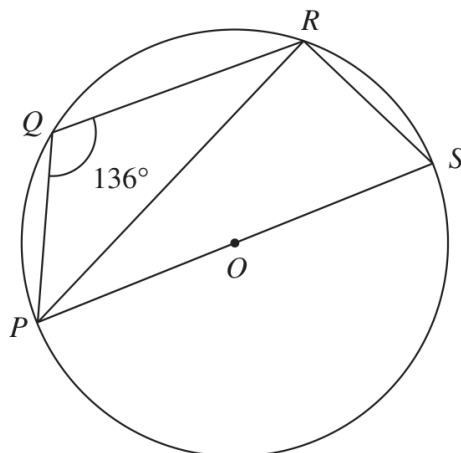
Angle $DFG = \dots\dots\dots^\circ$

(Total for Question 16 is 4 marks)

Q14 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1HR · Q15

CH 4 · TOPIC 4 3 marks

15

Diagram **NOT**
accurately drawn P , Q , R and S are points on a circle with centre O PS is a diameter of the circle.Angle $PQR = 136^\circ$ Work out the size of angle RPS

○

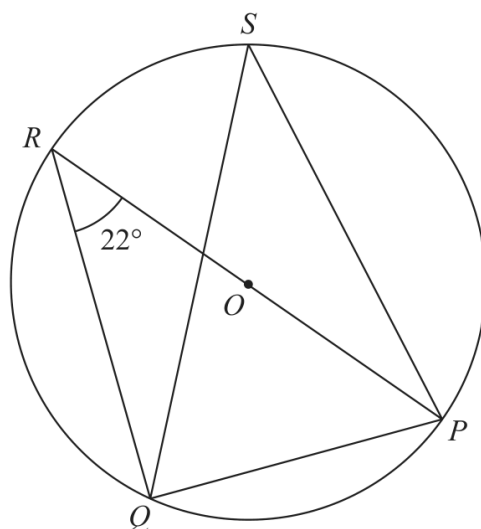
(Total for Question 15 is 3 marks)

Q15**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2H · Q14

CH 4 · TOPIC 4 4 marks

14

Diagram **NOT**
accurately drawn

P, Q, R and S are points on a circle, centre O
 ROP is a diameter of the circle.
 Angle $PRQ = 22^\circ$

(a) (i) Find the size of angle RQP

○

(ii) Give a reason for your answer.

(b) (i) Find the size of angle PSQ

○

(ii) Give a reason for your answer.

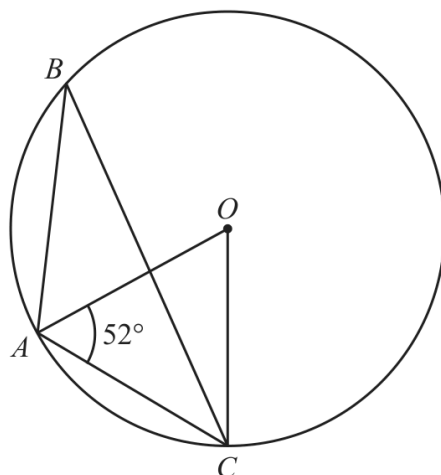
(Total for Question 14 is 4 marks)

Q16**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q14

CH 4 · TOPIC 4 3 marks

14

Diagram **NOT**
accurately drawn A , B and C are points on a circle, centre O Angle $OAC = 52^\circ$ Find the size of angle ABC

Give reasons for your working.

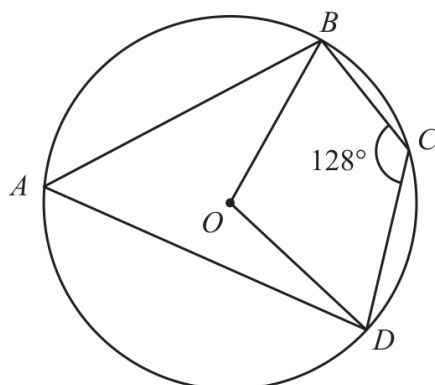
.....^o

(Total for Question 14 is 3 marks)

Q17 PAST PAPER Edexcel 4MA1 • May 2024 • Paper 1HR • Q13

CH 4 • TOPIC 4 5 marks

13

Diagram **NOT**
accurately drawn A , B , C and D are points on a circle, centre O Angle $BCD = 128^\circ$ Work out the size of angle OBD

Give a reason for each stage of your working.

angle $OBD = \dots\dots\dots^\circ$

(Total for Question 13 is 5 marks)

Q18**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 2H · Q16

CH 4 · TOPIC 4 5 marks

16 A, B, C and D are points on a circle, centre O

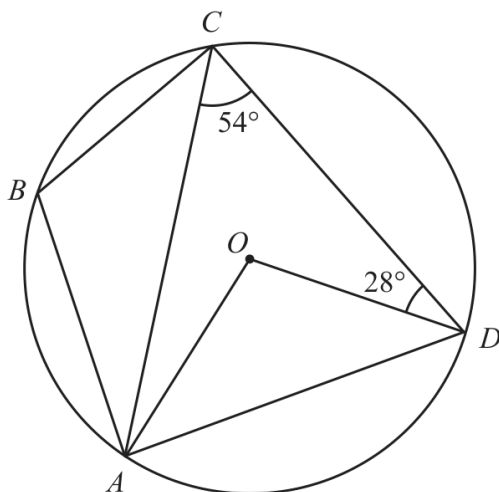


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

o

(ii) Give a reason for your answer to part (a)(i)

(b) Work out the size of angle CAO

o

(c) Work out the size of angle ABC

o

(Total for Question 16 is 5 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q15

CH 4 · TOPIC 4 3 marks

15 A , B and C are points on a circle, centre O

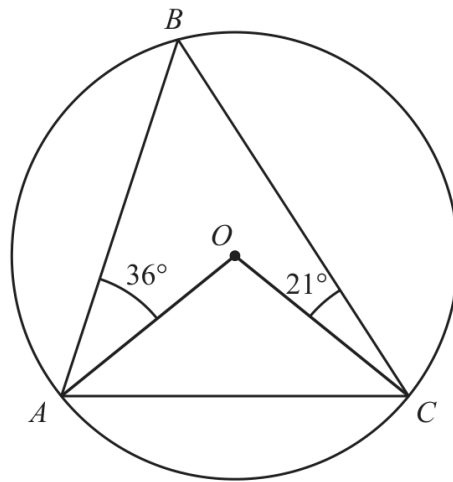


Diagram **NOT**
accurately drawn

Angle $BAO = 36^\circ$

Angle $BCO = 21^\circ$

Work out the size of angle ACO

.....^o

(Total for Question 15 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q15

CH 4 · TOPIC 4 4 marks

15 A, B, C and D are points on a circle with centre O

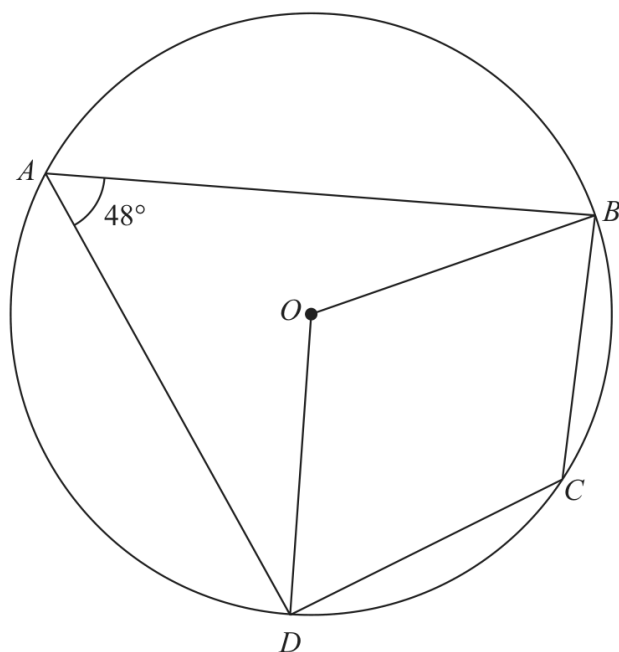


Diagram **NOT**
accurately drawn

Angle $DAB = 48^\circ$

(a) (i) Work out the size of the obtuse angle DOB

○

(ii) Give a reason for your answer.

(b) (i) Work out the size of angle BCD

○

(ii) Give a reason for your answer.

(Total for Question 15 is 4 marks)

02 Tangents

A tangent meets the radius at 90° . Two tangents from one point are equal. The angle between a tangent and a chord equals the angle in the alternate segment.

WHEN YOU SEE → YOU DO

“tangent at A” and O

→ $OA \perp$ tangent: 90°

two tangents from P

→ equal lengths; kite $OAPB$: $O + P = 180^\circ$

tangent and a chord at the same point

→ alternate segment theorem

“MN is the tangent at C” and a triangle in the circle

→ angle between = angle opposite in the triangle

WORKED MODEL • JUNE 2023 P2HR

MN is the tangent at C. $AB = CB$, angle $BCN = 78^\circ$. Find angle ABC.

1 $BAC = 78^\circ$

alternate segment theorem

2 $BCA = 78^\circ$

isosceles triangle ($AB = CB$)

ANSWER $ABC = 180 - 156 = 24^\circ$

IDEA 02

Practise: Tangents

My questions first, then selected practice, then real past papers.

Q21 DR ESLAM Two tangents

CH 4 • TOPIC 4 3 marks

Tangents from P touch the circle, centre O , at A and B . Angle $APB = 40^\circ$. Find angle AOB .

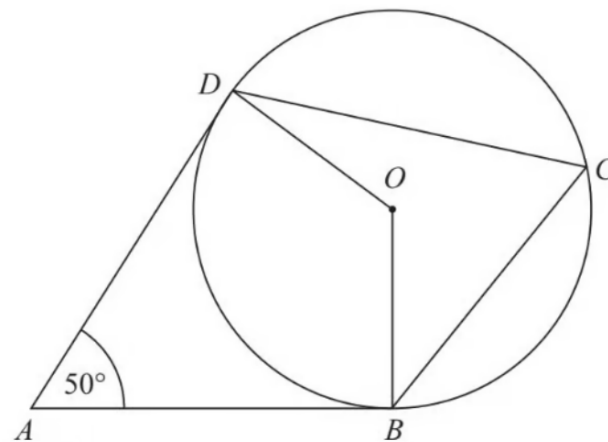
SELECTED PRACTICE

Q22**PRACTICE**

Practice · Medium

CH 4 · TOPIC 4 4 marks

1

Diagram **NOT**
accurately drawn

B , C and D are points on the circumference of a circle, centre O .
 AB and AD are tangents to the circle.

Angle $DAB = 50^\circ$

Work out the size of angle BCD .
 Give a reason for each stage in your working.

(4 marks)

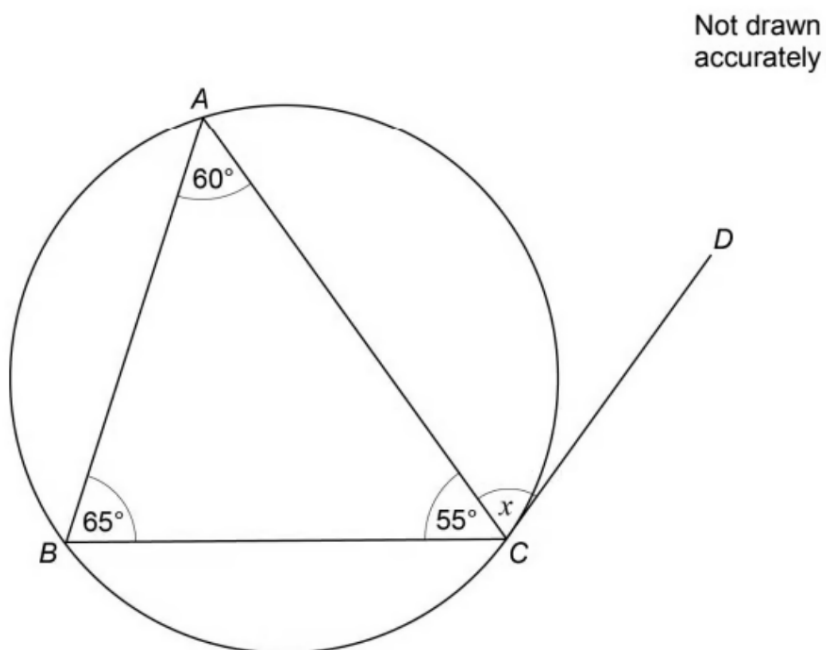
Q23**PRACTICE**

Practice · Medium

CH 4 · TOPIC 4 2 marks

16 A , B and C are points on a circle.

CD is a tangent to the circle.



Write down the size of angle x .

Give a reason for your answer.

.....degrees

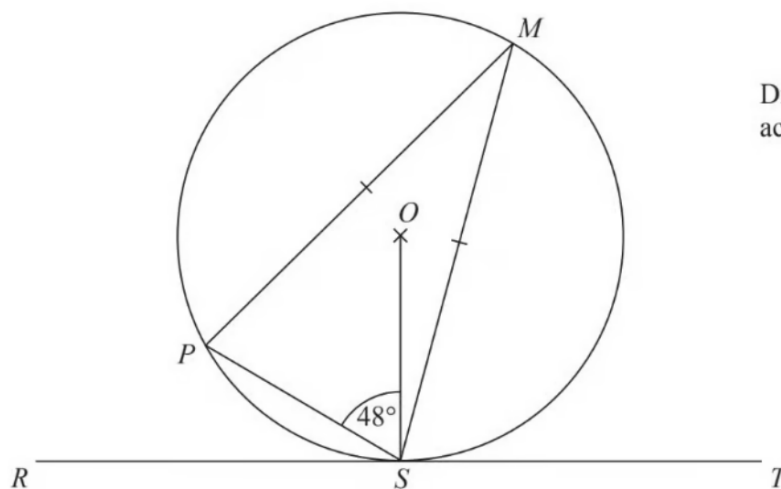
(2 marks)

Q24**PRACTICE**

Practice · Hard

CH 4 · TOPIC 4 5 marks

3

Diagram **NOT**
accurately drawn

P , M and S are points on a circle, centre O . RST is a tangent to the circle.

Angle $PSO = 48^\circ$

$MP = MS$

Work out the size of angle MST .

Give reasons for each stage of your working.

(5 marks)

Q25 PRACTICE Practice · Very Hard

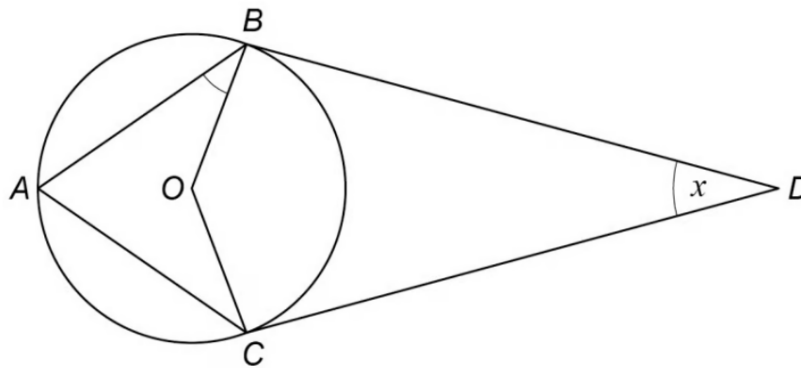
CH 4 · TOPIC 4 4 marks

12 A , B and C are three points on the circumference of a circle, centre O .

BD and CD are tangents to the circle.

$ABDC$ is a kite.

Angle BDC is x



Not drawn
accurately

Prove that angle ABO is $45^\circ - \frac{x}{4}$

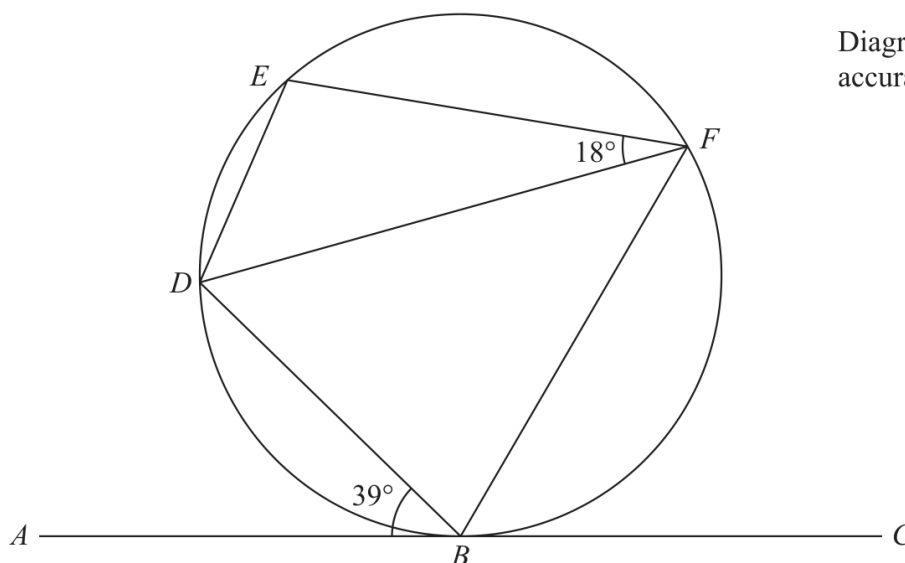
(4 marks)

PAST PAPERS

Q26 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 1HR · Q14

CH 4 · TOPIC 4 4 marks

14

Diagram **NOT**
accurately drawn

B, D, E and F are points on a circle.

ABC is the tangent at B to the circle.

Angle $ABD = 39^\circ$

Angle $EFD = 18^\circ$

Work out the size of angle BDE .

Give reasons for your working.

○

(Total for Question 14 is 4 marks)

Q27 PAST PAPER Edexcel 4MA1 • May 2021 • Paper 1H • Q15

CH 4 • TOPIC 4 4 marks

- 15 P , Q and R are points on a circle, centre O .
 TRV is the tangent to the circle at R .

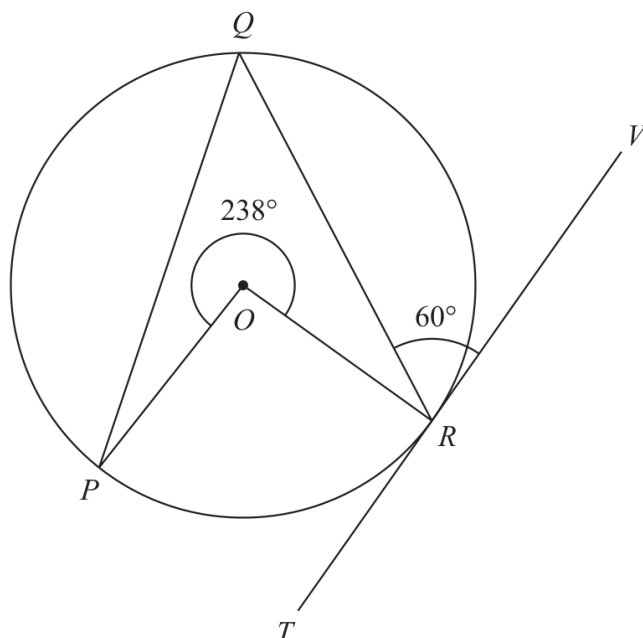


Diagram **NOT**
 accurately drawn

Reflex angle $POR = 238^\circ$

Angle $QRV = 60^\circ$

Calculate the size of angle OPQ .

Give a reason for each stage of your working.

o

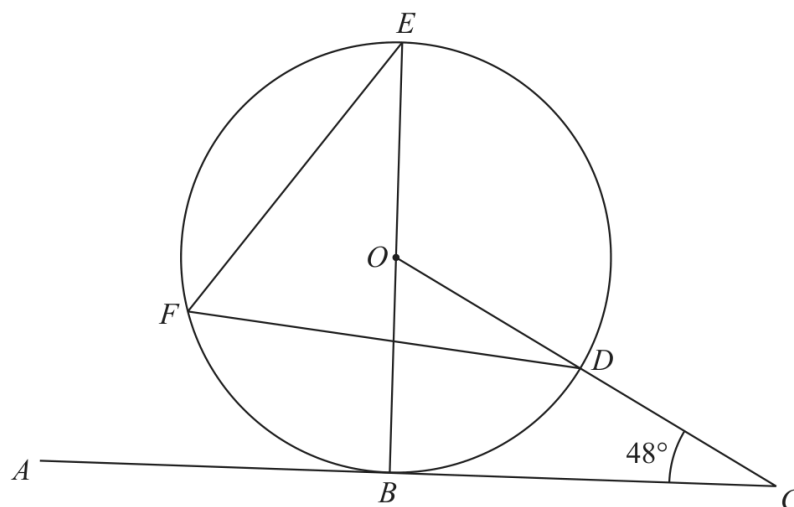
(Total for Question 15 is 4 marks)

Q28**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 2H · Q11

CH 4 · TOPIC 4 3 marks

11

Diagram **NOT**
accurately drawn

B, D, E and F are points on a circle, centre O .
 ABC is a tangent to the circle.
 ODC is a straight line.

BOE is a diameter of the circle.

Angle $BCD = 48^\circ$

Find the size of angle DFE .

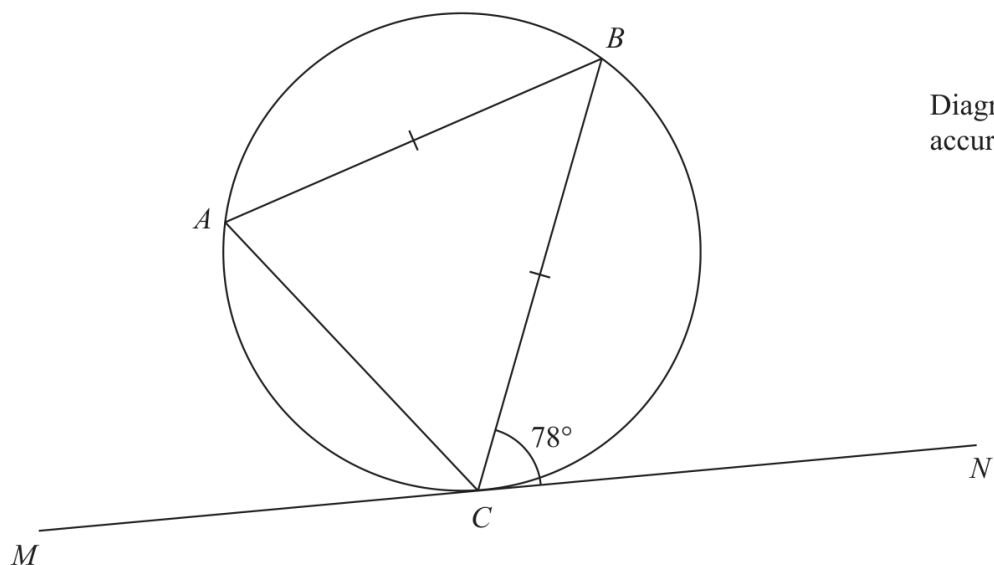
(Total for Question 11 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q20

CH 4 • TOPIC 4 2 marks

20 A , B and C are points on a circle.



MN is the tangent to the circle at C

$AB = CB$

Angle $BCN = 78^\circ$

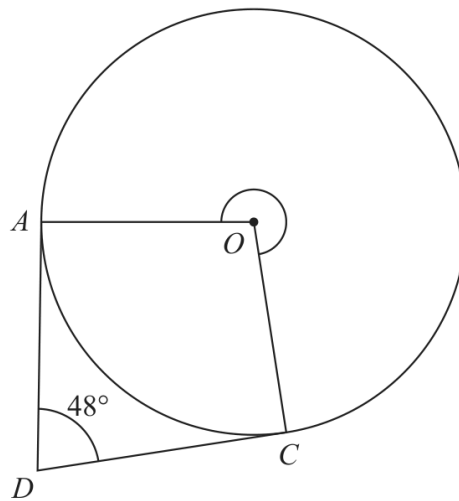
Find the size of angle ABC

(Total for Question 20 is 2 marks)

Q30 PAST PAPER Edexcel 4MA1 · June 2023 · Paper 2H · Q7

CH 4 · TOPIC 4 3 marks

7

Diagram **NOT**
accurately drawn

A and C are points on a circle, centre O
 DA is the tangent to the circle at A and DC is the tangent to the circle at C

Angle $ADC = 48^\circ$

Work out the size of reflex angle AOC

o

(Total for Question 7 is 3 marks)

03 Chords and Secants

Two chords crossing inside: part $\times$ part = part $\times$ part. Two lines from a point outside: outside part $\times$ whole line.

WHEN YOU SEE $\rightarrow$ YOU DO

chords APB and CPD cross at P

$$\rightarrow AP \times PB = CP \times PD$$

lines ABP and DCP from P outside

$$\rightarrow PB \times PA = PC \times PD$$

a tangent PC and a line PBA

$$\rightarrow PC^2 = PB \times PA$$

a diameter through the crossing

$\rightarrow$ use it as one chord, then the radius

ZONE INSIDE

$$AP \cdot PB = CP \cdot PD$$

parts of the chords

ZONE OUTSIDE

$$PB \cdot PA = PC \cdot PD$$

outside part $\times$ whole

ZONE TANGENT

$$PC^2 = PB \cdot PA$$

tangent squared

WORKED MODEL • JUNE 2023 P2HR

ABP and DCP are straight. $AB = 16$, $BP = 14$, $CP = 12$. Find DC .

1 $14 \times 30 = 12 \times PD$

outside $\times$ whole

2 $PD = 35$

solve

ANSWER $DC = 35 - 12 = 23 \text{ cm}$

$\times$ **TRAP** Using AB (the inside part) instead of the whole line $PA = 30$.

IDEA 03

Practise: Chords and Secants

My questions first, then selected practice, then real past papers.

Q31 DR ESLAM Intersecting chords

CH 4 · TOPIC 4 2 marks

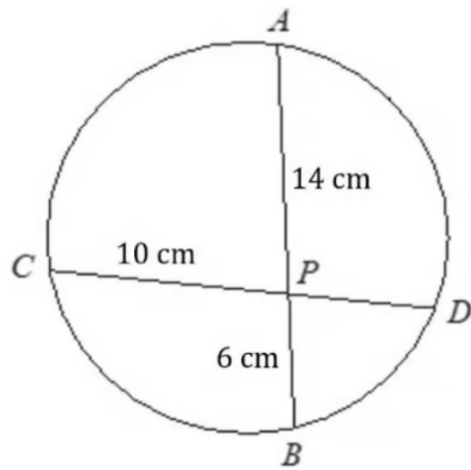
Chords AB and CD cross at P . $AP = 6$, $PB = 4$, $CP = 3$. Find PD .

SELECTED PRACTICE

Q32 PRACTICE Practice • Medium

CH 4 • TOPIC 4 2 marks

7

Diagram **NOT**
accurately drawn

APB and CPD are chords of a circle.

$AP = 14 \text{ cm}$ $PB = 6 \text{ cm}$ $CP = 10 \text{ cm}$

Calculate the length of PD .

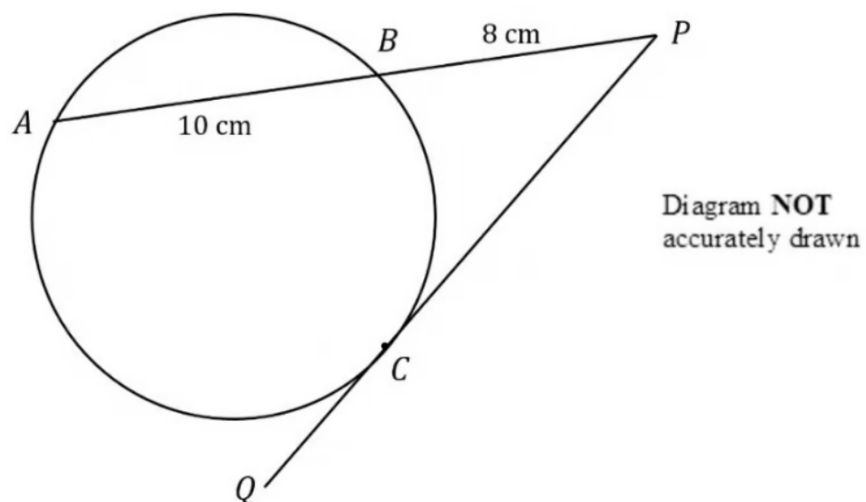
(2 marks)

Q33**PRACTICE**

Practice · Hard

CH 4 · TOPIC 4 3 marks

7



A , B and C are points on a circle.

ABP is a straight line.

QCP is a tangent to the circle.

$AB = 10 \text{ cm}$

$BP = 8 \text{ cm}$

Work out the length of CP .

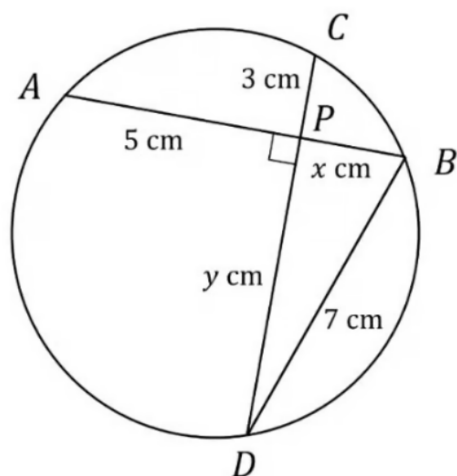
(3 marks)

Q34**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 4 6 marks

4

Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle.
 APB , CPD , and BD are chords of the circle.
 APB and CPD intersect perpendicularly at P .

$BD = 7 \text{ cm},$	$AP = 5 \text{ cm},$	$CP = 3 \text{ cm},$	$PB = x \text{ cm},$	and	$PD = y \text{ cm}.$
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Work out the values of x and y .
 Show your working clearly.
 Give your answers correct to 3 significant figures.

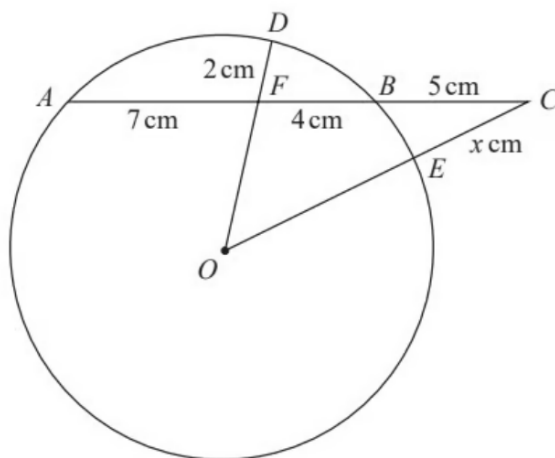
(6 marks)

Q35**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 4 6 marks

9

Diagram NOT
accurately drawn

A , D , B and E are points on a circle, centre O .

$AFBC$, OEC and OFD are straight lines.

$AF = 7$ cm, $FB = 4$ cm, $BC = 5$ cm, $FD = 2$ cm and $CE = x$ cm.

Work out the value of x .

Show your working clearly.

$x = \dots\dots\dots$

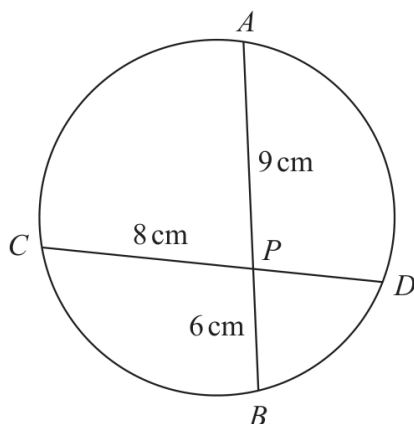
(6 marks)

PAST PAPERS

Q36 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 2HR • Q18

CH 4 • TOPIC 4 2 marks

18

Diagram **NOT**
accurately drawn APB and CPD are chords of a circle. $AP = 9\text{ cm}$ $PB = 6\text{ cm}$ $CP = 8\text{ cm}$ Calculate the length of PD .

..... cm

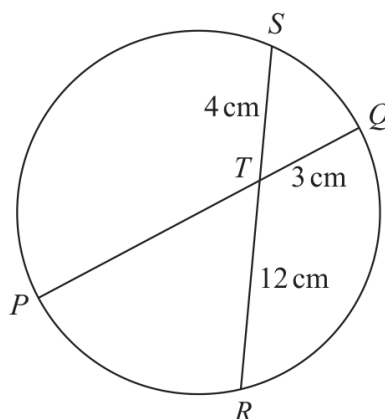
(Total for Question 18 is 2 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q19

CH 4 • TOPIC 4 3 marks

19

Diagram **NOT**
accurately drawn

PTQ is a diameter of a circle.
 RTS is a chord of the circle.

$$TQ = 3 \text{ cm} \qquad ST = 4 \text{ cm} \qquad TR = 12 \text{ cm}$$

Calculate the radius of the circle.

..... cm

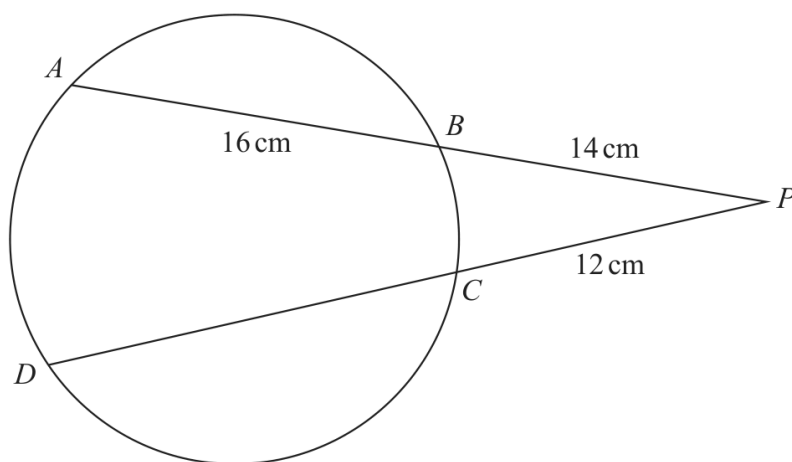
(Total for Question 19 is 3 marks)

Q38**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2HR · Q18

CH 4 · TOPIC 4 3 marks

18

Diagram **NOT**
accurately drawn A, B, C and D are points on a circle. ABP and DCP are straight lines.

$AB = 16 \text{ cm}$

$BP = 14 \text{ cm}$

$CP = 12 \text{ cm}$

Work out the length of DC

..... cm

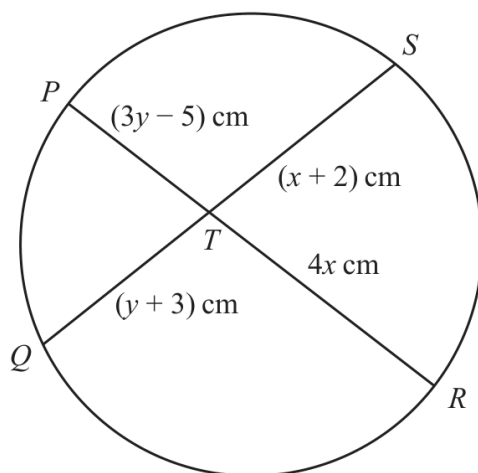
(Total for Question 18 is 3 marks)

Q39**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q23

CH 4 · TOPIC 4 5 marks

23

Diagram **NOT**
accurately drawn PTR and QTS are chords of a circle.

$$PT = (3y - 5) \text{ cm} \quad QT = (y + 3) \text{ cm} \quad RT = 4x \text{ cm} \quad ST = (x + 2) \text{ cm}$$

Find an expression for y in terms of x

$$y = \dots\dots\dots$$

Total for Question 23 is 5 marks)

04 Proofs with Circle Theorems

Write the proof as a chain: one statement, one reason, per line. End with the sentence you were asked to prove.

WHEN YOU SEE → YOU DO

“prove the triangles are similar”

→ two pairs of equal angles, with reasons

“prove AB is parallel to DC ”

→ find equal alternate (or corresponding) angles

“prove that angle $CDA = 2x$ ”

→ follow the angles round, naming each theorem

“prove that QS is not a diameter”

→ show an angle that should be 90° is not

WORKED MODEL • MAY/NOV 2020 P1HR

Tangent ABC at B ; angle $EDF = 40^\circ$, angle $FBC = 70^\circ$. Prove $ABC \parallel EF$.

1 $FDB = 70^\circ$

alternate segment theorem

2 $EDB = 110^\circ \Rightarrow EFB = 70^\circ$

opposite angles of a cyclic quadrilateral

ANSWER $EFB = FBC = 70^\circ$: alternate angles, so $ABC \parallel EF$

IDEA 04

Practise: Proofs with Circle Theorems

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE

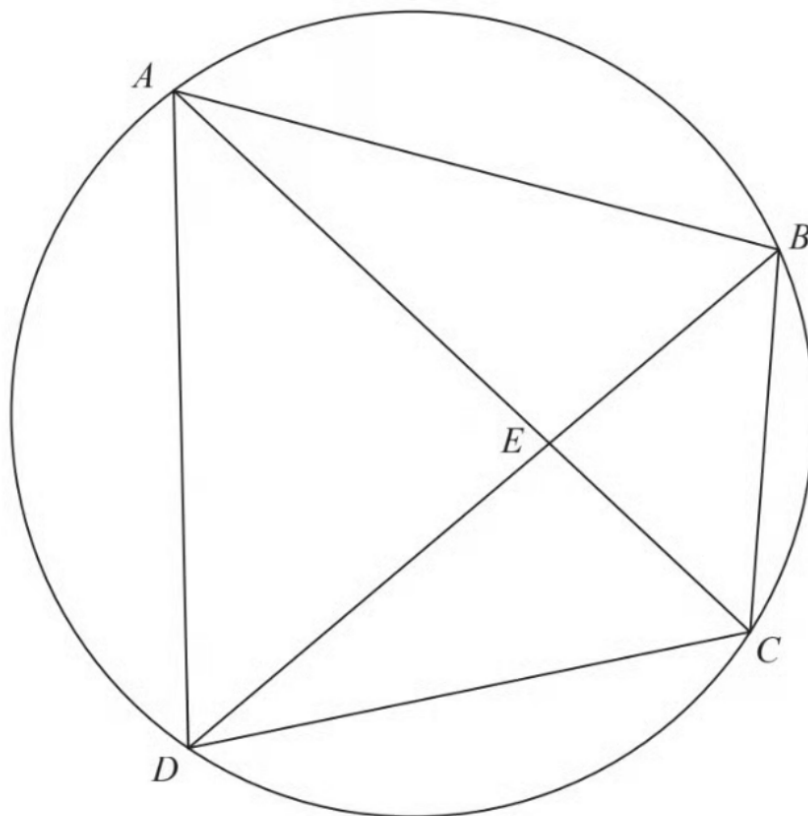
Q40

PRACTICE

Practice · Hard

CH 4 · TOPIC 4 3 marks

6 A , B , C and D are four points on the circumference of a circle.



AEC and BED are straight lines.

Prove that triangle ABE and triangle DCE are similar.

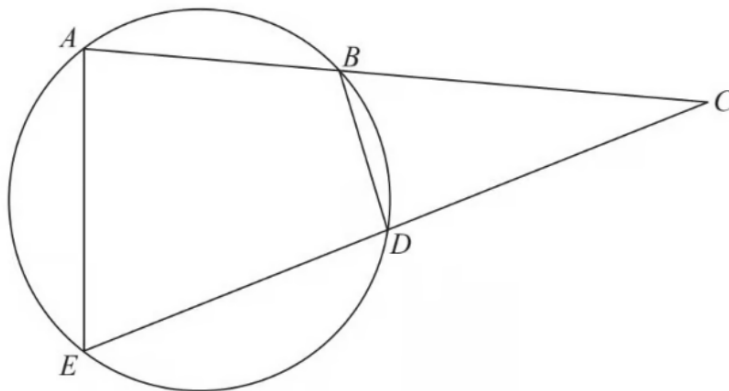
You must give reasons for each stage of your working.

(3 marks)

Q41 PRACTICE Practice · Very Hard

CH 4 · TOPIC 4 5 marks

1

Diagram NOT
accurately drawn

A , B , D and E are points on a circle.

ABC and EDC are straight lines.

Prove that triangle BCD is similar to triangle ECA .

You must give reasons for your working.

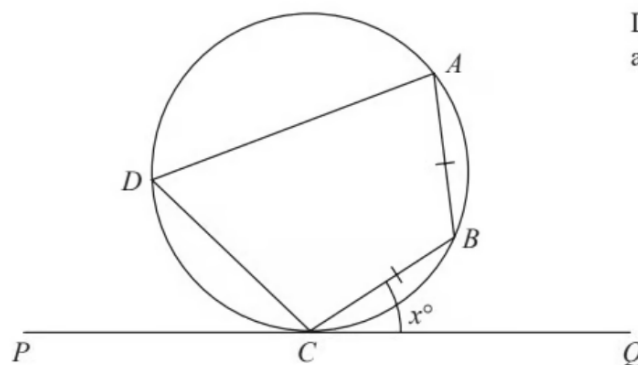
(5 marks)

Q42**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 4 5 marks

7

Diagram **NOT**
accurately drawn A, B, C and D are points on a circle. PCQ is a tangent to the circle. $AB = CB$.Angle $BCQ = x^\circ$ Prove that angle $CDA = 2x^\circ$

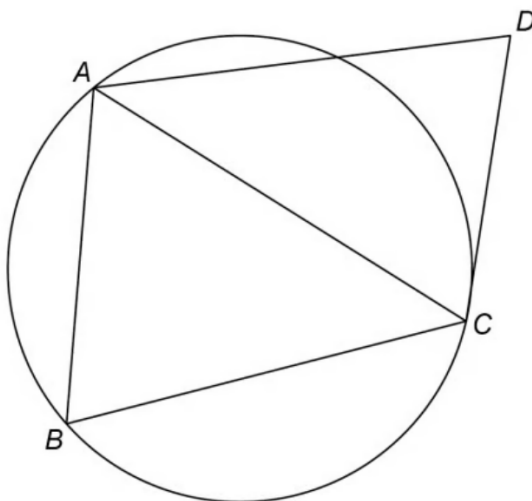
Give reasons for each stage in your working.

(5 marks)

Q43**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 4 5 marks

14 (a) A , B and C are points on a circle. CD is a tangent.Not drawn
accuratelyAssume that triangle ABC is isosceles with $AC = BC$ Prove that AB is parallel to DC .

(4 marks)

(b) In fact, triangle ABC is equilateral.Tick the **two** boxes for the statements that **must** be correct.

☐	AB is parallel to DC
☐	AC bisects angle BCD
☐	AC bisects angle BAD

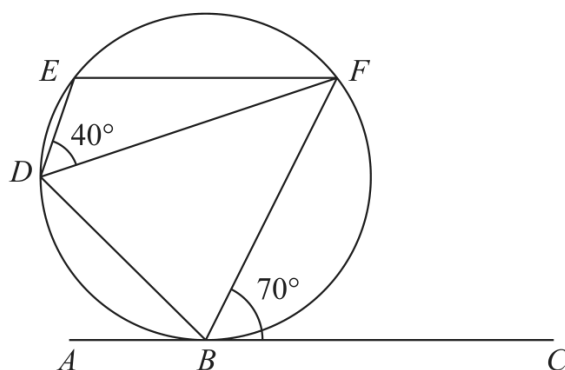
(1 mark)

PAST PAPERS

Q44 PAST PAPER Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q15

CH 4 · TOPIC 4 4 marks

15

Diagram **NOT**
accurately drawn

B , D , E and F are points on a circle.
 ABC is the tangent to the circle at B .

Angle $EDF = 40^\circ$

Angle $FBC = 70^\circ$

Prove that the tangent ABC is parallel to EF .
Give a reason for each stage of your working.

(Total for Question 15 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------|--|
| 01 | Angles | Centre $2\times$; semicircle 90 ; same segment; cyclic 180 ; radii isosceles. |
| 02 | Tangents | $\perp$ radius; two tangents equal; alternate segment. |
| 03 | Chords | Inside: part·part; outside: outside·whole; tangent squared. |
| 04 | Proofs | One statement + one reason per line. |

BOTTOM LINE Spot the shape → name the theorem → one angle at a time.

Answers

Circle Theorems · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · ANGLES IN CIRCLES

- Q1** Dr Eslam • Centre and circumference
94°: the angle at the centre is twice the angle at the circumference
- Q2** Dr Eslam • Cyclic quadrilateral
107°: opposite angles of a cyclic quadrilateral add to 180°
- Q3** Practice • Medium
 $ADC = \frac{y}{2}$ (centre = 2 × circumference): $ABC = 180 - \frac{y}{2}$ (cyclic quadrilateral)
- Q4** Practice • Medium
 AD : in triangle ACD , $ACD = 180 - 10 - 80 = 90^\circ$ (angle in a semicircle)
- Q5** Practice • Hard
(a) $ACD = 58^\circ$ (angles in the same segment)
(b) $ABC = 90^\circ$, $BAC = 22^\circ$: $ACB = 68^\circ$
- Q6** Practice • Very Hard
 $BCD = 90^\circ$, $CBD = 28^\circ$: $BD = \frac{6.4}{\sin 28^\circ} = 13.6$: area 146 cm^2
 $AOC = 76^\circ$: $OAC = 52^\circ$
- Q7** Edexcel 4MA1 • January 2020 • Paper 1HR • Q14
 $ABD = 90^\circ$, $BAD = 58^\circ$, $BCD = 122^\circ$: $BDC = 30^\circ$
- Q8** Edexcel 4MA1 • January 2020 • Paper 2H • Q13
(a) 122° , opposite angles of a cyclic quadrilateral (b) 244°
- Q9** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q16
(a)(i) $m = 62$ (ii) $n = 118$ (b) 62°
- Q10** Edexcel 4MA1 • January 2021 • Paper 2H • Q13
 $AC = 11.70$; $ADC = 82^\circ$; sine rule: $AD = 6.78$, $CD = 10.53$: perimeter 32.8 cm
- Q11** Edexcel 4MA1 • May 2021 • Paper 2H • Q18
 $ABC = 90^\circ$, $ACB = 35^\circ$: $ADB = 35^\circ$
- Q12** Edexcel 4MA1 • January 2022 • Paper 1HR • Q14
 $EFG = 90^\circ$, $DFE = 42^\circ$: $DFG = 48^\circ$
- Q13** Edexcel 4MA1 • January 2022 • Paper 2H • Q16
 $PSR = 44^\circ$, $PRS = 90^\circ$: $RPS = 46^\circ$
- Q14** Edexcel 4MA1 • June 2022 • Paper 1HR • Q15
(a) 90° , angle in a semicircle (b) 22° , angles in the same segment
- Q15** Edexcel 4MA1 • June 2023 • Paper 2H • Q14
 $AOC = 76^\circ$: $ABC = 38^\circ$
- Q16** Edexcel 4MA1 • November 2023 • Paper 1H • Q14
 $BAD = 52^\circ$, $BOD = 104^\circ$: $OBD = 38^\circ$
- Q17** Edexcel 4MA1 • May 2024 • Pa-

IDEA 02 • TANGENTS

- Q21** Dr Eslam • Two tangents $360 - 90 - 90 - 40 = 140^\circ$
- Q22** Practice • Medium $OBA = ODA = 90^\circ, BOD = 130^\circ: BCD = 65^\circ$
- Q23** Practice • Medium $x = 65^\circ$ (alternate segment theorem)
- Q24** Practice • Hard $PSR = 42^\circ$ (tangent $\perp$ radius), $PMS = 42^\circ$ (alternate segment), $MSP = 69^\circ: MST = 69^\circ$
- Q25** Practice • Very Hard $BOC = 180 - x; BAC = 90 - \frac{x}{2}$; kite symmetric about AO: $ABO = BAO = 45 - \frac{x}{4}$
- Q26** Edexcel 4MA1 • January 2021 • Paper 1HR • Q14 $BFD = 39^\circ$ (alternate segment), $EFB = 57^\circ: BDE = 123^\circ$ (cyclic quadrilateral)
- Q27** Edexcel 4MA1 • May 2021 • Paper 1H • Q15 $PQR = 61^\circ, OQR = ORQ = 30^\circ: OPQ = 31^\circ$
- Q28** Edexcel 4MA1 • November 2021 • Paper 2H • Q11 $BOD = 42^\circ, BED = 21^\circ, DBE = 69^\circ: DFE = 69^\circ$
- Q29** Edexcel 4MA1 • June 2023 • Paper 2HR • Q20 $BAC = 78^\circ$ (alternate segment) = $BCA: ABC = 24^\circ$
- Q30** Edexcel 4MA1 • June 2023 • Paper 2H • Q7 $AOC = 132^\circ: \text{reflex } 228^\circ$

IDEA 03 • CHORDS AND SECANTS

- Q31** Dr Eslam • Intersecting chords
 $6 \times 4 = 3 \times PD: PD = 8$
- Q32** Practice • Medium
 $14 \times 6 = 10 \times PD: PD = 8.4 \text{ cm}$
- Q33** Practice • Hard
 $CP^2 = 8 \times 18 = 144: CP = 12 \text{ cm}$
- Q34** Practice • Very Hard
 $5x = 3y \text{ and } x^2 + y^2 = 49: x = 3.60, y = 6.00$
- Q35** Practice • Very Hard
 $AF \cdot FB = DF(2r - DF): r = 8; x(x + 16) = 80:$
 $x = 4$
 $PD = 6.75 \text{ cm}$
- Q36** Edexcel 4MA1 • January 2020 • Paper 2HR • Q18
- Q37** Edexcel 4MA1 • January 2022 • Paper 2H • Q19
 $12 \times 4 = 3 \times PT, PT = 16: \text{radius } 9.5 \text{ cm}$
- Q38** Edexcel 4MA1 • June 2023 • Paper 2HR • Q18
 $14 \times 30 = 12 \times PD: DC = 23 \text{ cm}$
- Q39** Edexcel 4MA1 • June 2025 • Paper 1HR • Q23
 $(3y - 5)(4x) = (y + 3)(x + 2): y = \frac{23x + 6}{11x - 2}$

IDEA 04 • PROOFS WITH CIRCLE THEOREMS

- Q40** Practice • Hard
 $BAC = BDC$ (same segment), $ABD = ACD$ (same segment), $AEB = DEC$: similar
- Q41** Practice • Very Hard
 $BCD = ECA$ (common), $CBD = CEA$ (exterior angle of cyclic quadrilateral): similar
- Q42** Practice • Very Hard
 $BAC = x$ (alternate segment) = BCA (isosceles); $ABC = 180 - 2x: CDA = 2x$ (cyclic)
- Q43** Practice • Very Hard
 (a) $ABC = BAC$ (isosceles) = ACD (alternate segment): alternate angles, so $AB \parallel DC$ (b) $AB \parallel DC$ and AC bisects angle BCD
- Q44** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q15
 $FDB = 70^\circ$ (alternate segment); $EDB = 110^\circ$, $EFB = 70^\circ$ (cyclic) = FBC : alternate angles, so parallel